

Fluidized Bed Lab Report

Unit Operation Lab #4 

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Group 2 of the Tuesday AM section

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1. Executive Summary

The objective of this experiment was to determine the minimum superficial velocity by analyzing the performance of a fluidized bed system with two different types of particles, silica sand and ion exchange resin as a function of pressure drop and bed height. Initially, the system was a static packed bed, which then transformed into a fluidized bed as the flow rate was increased. The minimum fluidization velocity was reached when the value of the pressure drops stopped changing. Fluidized beds are essential in process technologies because of their uniform heat and mass transfer.

The pressure drop was measured while increasing and decreasing the system's flow rate with 20 liters per minute (LPM) intervals. The pressure changes were manipulated by varying flow rates ranging from 20 LPM to 100 LPM at six and ten-inch bed heights. The experimental values were compared to the theoretical calculations obtained using the Ergun equation to calculate the minimum velocity and evaluate the percent error of the pressure drops. The porosities of silica sand and the ion exchange resin were 0.49 and 0.52, respectively. These values were found experimentally and calculated with volume changes. Five particle measurements were taken to calculate the average particle size of 257.81 μm . These values were required to apply the Ergun equation.

After all the experimental data was evaluated, the minimum fluidization velocity at the 6-inch and 10-inch bed height for the silica sand and ion exchange was 0.219 m/s. The actual velocity was moderately close to the theoretically calculated velocity for the ion exchange resin at a velocity of 0.271 m/s yielding a 19% error. There was an increased deviation in comparing the silica sand's actual minimum fluidization velocity with the theoretical one of 0.169 m/s. The experimental velocity was higher which caused a 30% error. Overall, the percent error for both types of particles was higher than expected because the flow rate was not uniformly controlled due to human error. Also, the system in the laboratory is worn and there are cracks at the top of the bed that allow for gases to escape.

2. Introduction

Semi-empirical correlations were developed to determine a minimum velocity of fluidization. In this experiment, the purpose was to determine the minimum superficial velocity and the correlation between pressure drop and the flow rate across a fluidized bed. Once the pressure vs. flowrate data is determined, that data was compared with the predicted data proving the validity of the correlations and the significance of this experiment.

Fluidized beds offer superior heat transfer, the ability to process materials with a wide particle size distribution and the ability to move solids like a fluid [2]. A few industrial applications include; drying, cooling, pneumatic transport of powders, coal gasification and separation of solids. In this experiment, the bed contains two small particles of solids. The bed particles consist of fine silica sand or ion exchange resin beads. The fluidized bed vessel consists of a 4-inch plexiglass tube enlarged at the top to reduce entrainment losses. At the bottom of the vessel, air is introduced through the control of valves [1].

3. Theory

A fluidized bed is a bed of small solid particles suspended and maintained in motion by an upward fluid flow [1]. Due to the fluid flow, a rapid mixture of the particles occurs. As the velocity increases, the pressure drop across the bed increases also. The pressure drop can be calculated by the Ergun equation:

$$\Delta P = \left(\frac{150 \bar{V}_0 \mu}{\Phi_s^2 D_p^2} * \frac{1 - \varepsilon^2}{\varepsilon^3} + \frac{1.75 \rho \bar{V}_0^2}{\Phi_s D_p} \frac{1 - \varepsilon}{\varepsilon^3} \right) * L$$

Where

ΔP is the pressure drop across the bed

$\bar{V}_0$ is the superficial velocity

μ is the fluid viscosity

ε is the bed voidage at minimum fluidization

ρ is the fluid density

D_p is the particle diameter

Φ_s is the sphericity

L is the bed height

Once the drag force of the upward moving gas is equal to the apparent weight of the particle, the minimum fluidization velocity is reached [3]. Referring to **Figure 3.1**, the minimum fluidization velocity is defined as point C. The minimum fluidization velocity for very small particles was compared to the calculated value from Ergun equation:

$$\bar{V}_{0M} \approx \frac{g(\rho_p - \rho)}{150\mu} \frac{\varepsilon_M^3}{1 - \varepsilon_M} \Phi_s^2 D_p^2$$

Where:

μ is the fluid viscosity

Φ_s is the sphericity

ε_M is the bed voidage at minimum fluidization

ρ_p is the particle density and ρ is the fluid density

The particle size of the sand is then calculated using:

$$D_{pav} = \frac{1}{\sum \left\{ \frac{x_i}{D_{pi}} \right\}}$$

Where

x_i is the mass fraction of particle of size D_{pi}

In **Figure 3.1**, the pressure drop across the bed counterbalances the force of gravity on the particles and the pressure drop reaches its maximum. When the velocity increases further, the particles separate slightly and begin to move in the bed which is described by point B. True fluidization begins and the pressure drop across the bed stays constant while the bed height increases with increasing flow [4]. The pressure drop differs for the superficial velocities before point C; the pressure drop is larger for the increasing flowrate compared to the pressure drop of the decreasing flowrate because the particles have to overcome the friction force as well as the drag force [5].

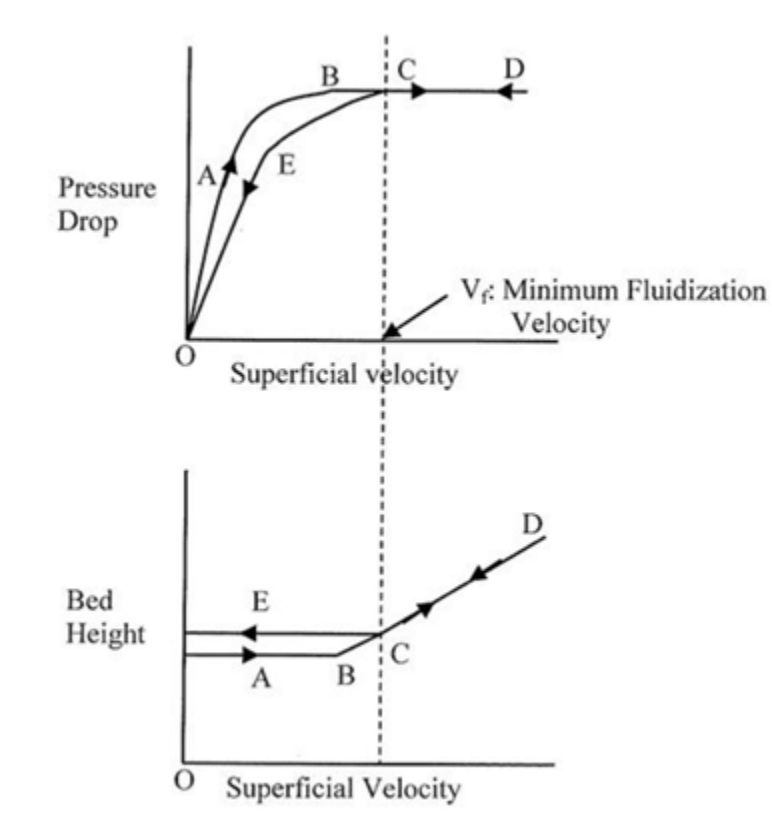


Figure 3.1: Pressure drop and bed height vs. superficial velocity [1]

4. Results

Two sets of experiments were performed where the mixture of fine silica sand is the bed material for the first experiment and ion exchange resin beads are for the second. For each material, the bed was packed to 6 and 10 inches respectively. The pressure drop was observed per unit bed height as a function of increasing and decreasing flowrates as shown in the figures below and Table 7.2, 7.4, 7.6 and 7.8 in Appendix I.

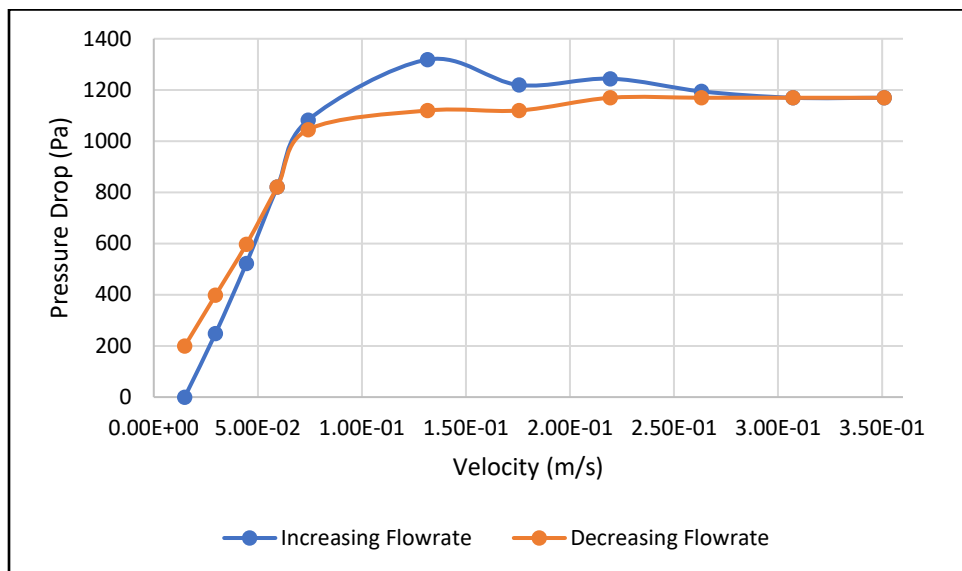


Figure 4.1: Velocity Vs. Pressure Drop of 6in. of Silica Sand

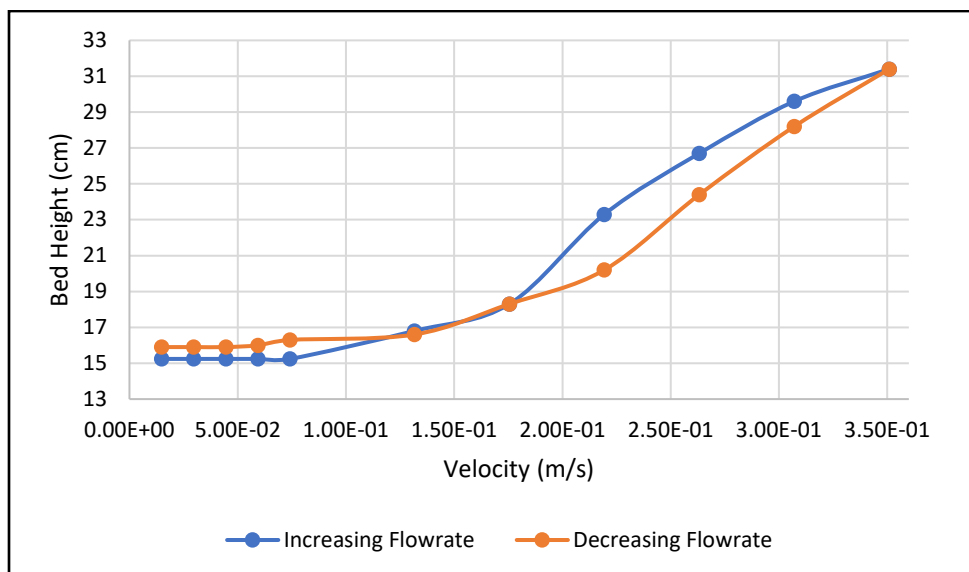


Figure 4.2: Velocity Vs. Bed Height of 6in. of Silica Sand

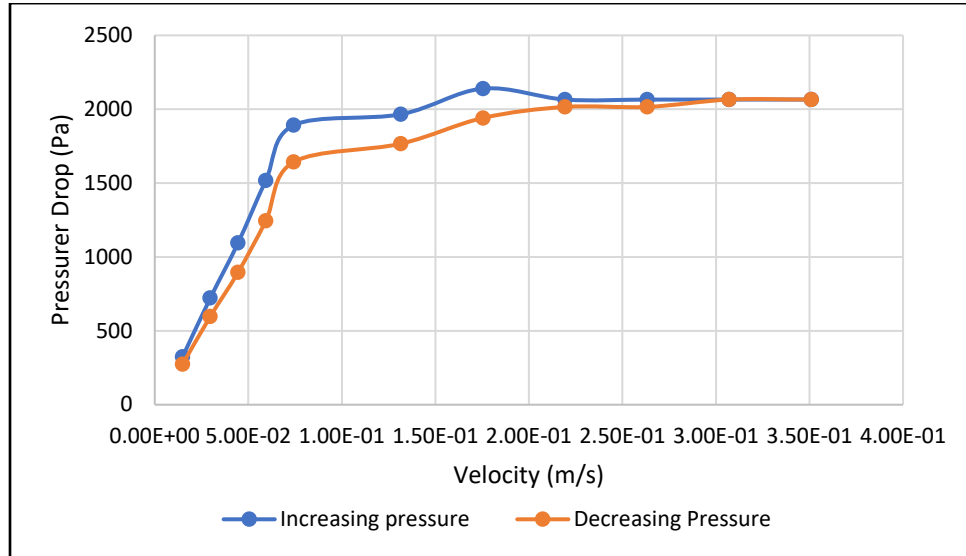


Figure 4.3: Velocity Vs. Pressure Drop of 10in. of Silica Sand

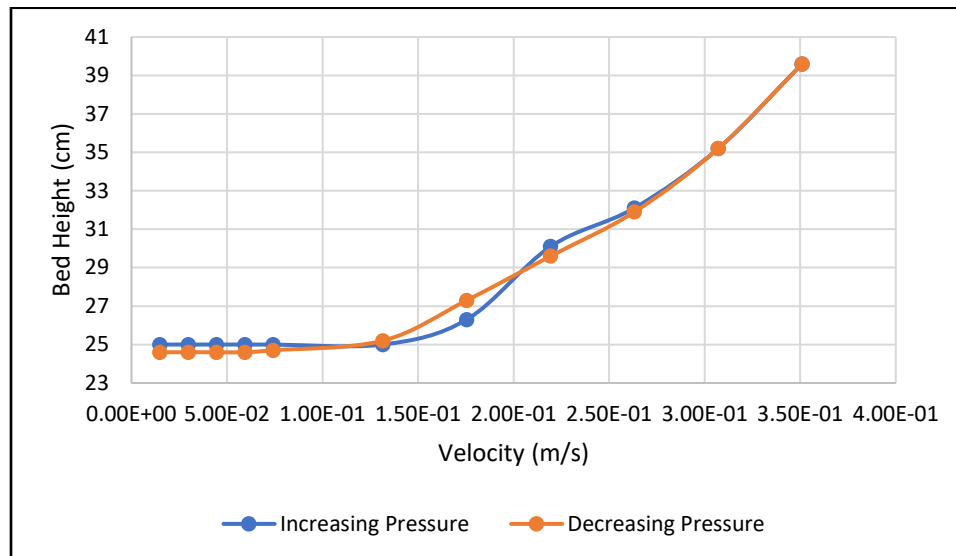


Figure 4.4: Velocity Vs. Bed Height of 10in. of Silica Sand

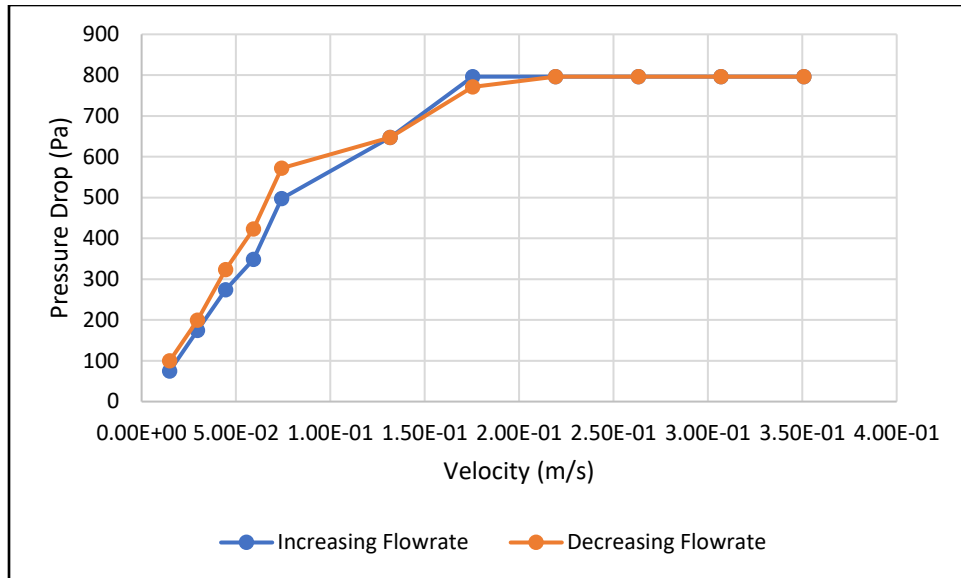


Figure 4.5: Velocity Vs. Pressure Drop of 6in. of Resin

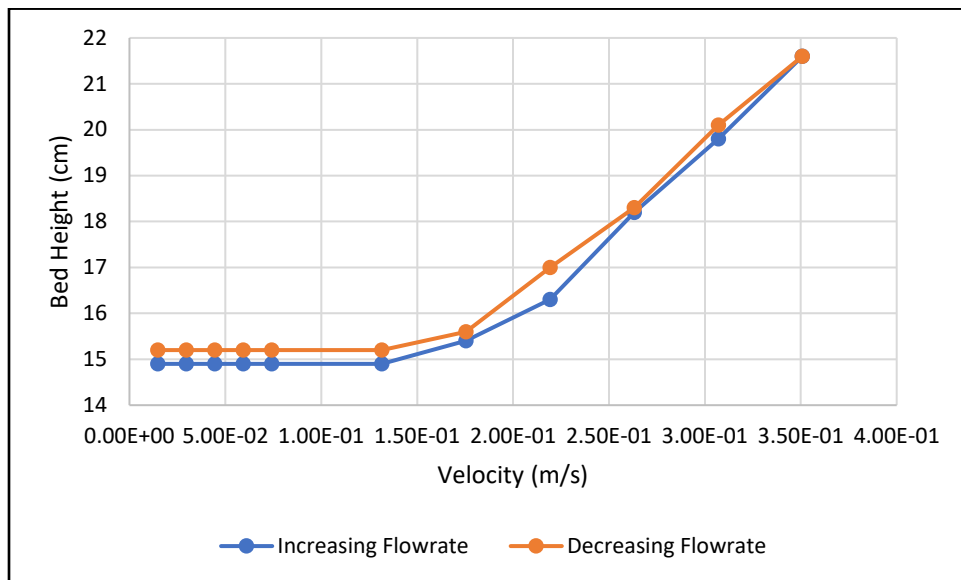


Figure 4.6: Velocity Vs. Bed Height of 6in. of Resin

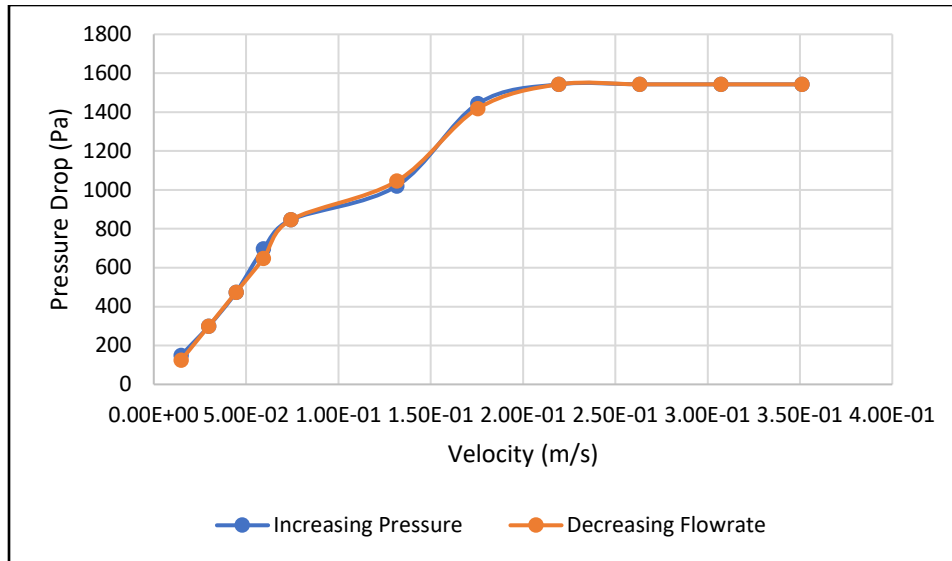


Figure 4.7: Velocity Vs. Pressure Drop of 10in. of Resin

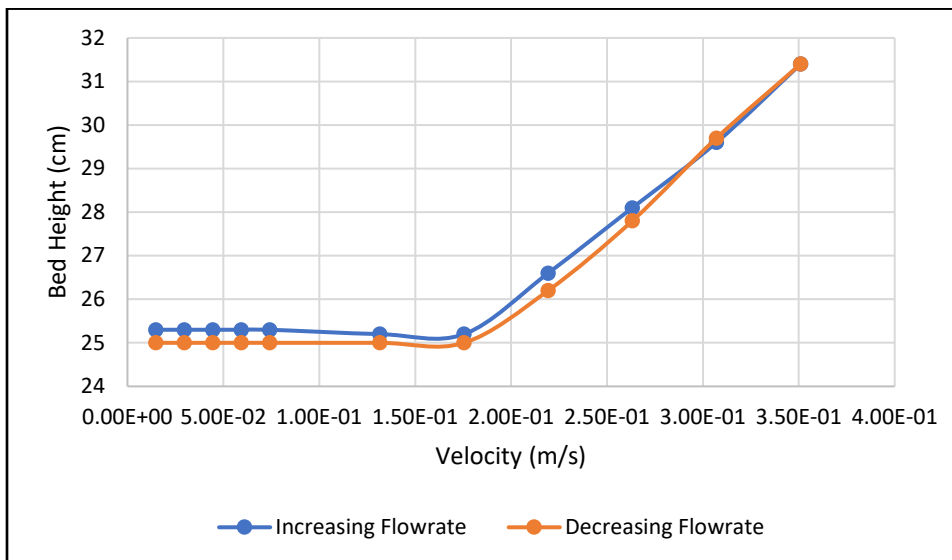


Figure 4.8: Velocity Vs. Bed Height of 10in. of Resin

For both experiments, it can be observed that as the flowrate increased, the pressure drop and bed height did also. When analyzing the graphs of velocity vs. pressure drop, a plateau is observed. This is due to no significant increase in pressure drop when the gas velocity is increased past the initial fluidization velocity and minimum fluidization velocity is reached [1].

5. Discussion

It can be shown in the results from the experimental data that the experiment agrees with most of the theory. However, two phenomena are not observed due to the lack of enough data points. The two phenomena are: the pressure drop for 6 in. and 10 in. silica sand being equal and the bed heights for 6 in. silica sand are equal for increased and decreased flow rates beyond the minimum fluidization velocity. Had the flow rate been continually increased past what was perceived as the minimum fluidization velocity, it could have been better illustrated that increasing or decreasing the flow rates would have had minimal or no effect on those parameters. An unexpected conclusion from the data is there was no discernable difference between the pressure drops for increasing and decreasing the flowrates for the 10 in. resin bed trial. With the increased bed height from 6 in. to 10 in., there is a similarity in the way the pressure drop changes with variable flow rates. However, the effect that increasing or decreasing the flow rates has on the pressure drop is minimized when considering both approaching and departing the minimum fluidization velocity at increased initial bed heights. This raised questions in how increased initial bed height minimizes these effects for both silica sand and resin beads. One way this could be analyzed is calculating the area between the two curves and comparing that between trials.

There are a few proposed changes in the experiment that would allow for more accurate results. Having two valves for varying the flow rate seems unnecessary and allows for more errors to arise when performing data extraction. The flow rate should have been increased continually from the beginning to the end of the data collection; however, due to the second valve, the flow rate was reduced to zero for a several seconds between switching valves. This may have affected the bed height and pressure drop in subsequent trials, for both increasing and decreasing flow rates. A single valve with a digital flow rate reader would have minimized these effects. The second change would be changing how the bed height is calculated. It was difficult to get an accurate reading on the bed height while the flow of air was passing through. At times estimates were over and under compensated, this meant a substantial error could have been introduced during data extraction. Using some sort of outlet for the sand or resin at specific heights would allow us to more accurately record the bed height. The floating ball for the right valve and the weight on the left valve should have been identical. Taking measurements from two different pieces of equipment produced inconsistencies in readings. The same vacuum was used to retrieve the sand and resin after experiment. This meant that there's a possibility that sand particles got into the resin particles and vice versa. Finally, the sand and resin were not new when we conducted the experiment. Previous groups may have further contaminated the sample we used for the experiment. We have no way of determining the degree of contamination.

Even with the limitations of both the experimental design, environmental factors, and the lack of enough data points, the experiment did produce results consistent with the theory. It might be beneficial to explore how the initial bed heights affects the change in bed heights for both increasing and decreasing flow rates.

6. **References**

- [1] Zdunek, Alan. “*Fluid Flow Through Fluidized Beds.*” ChE 381: Unit Operations Laboratory. Summer 2014. University of Illinois at Chicago Chemical Engineering Department
- [2] Cocco, R.; Karri, S.B.R.; Knowlton, T. “*Introduction to Fluidization.*” American Institute of Chemical Engineers. Nov. 2014.
- [3] Anantharaman, A.; Cocco, R.; Chew, J.W. “*Evaluation of correlations for minimum fluidization velocity (U_{mf}) in gas-solid fluidization.*” Science Direct. Powder Technology. Volume 323. January 2018. DOI: doi.org/10.1016/j.powtec.2017.10.016
- [4] McCabe, W.; Smith, J.; Harriot, P. *Unit Operations of Chemical Engineering.* 7th ed. McGraw-Hill, 2005.
- [5] Bird, B.R.; Stewart, W.E.; Lightfoot E.N.; *Transport Phenomena.* 2nd ed. John Wiley & Sons, Inc., 2002

7. Appendix I: **Data** Tabulation/Graphs

Table 7.1: Sand Properties

Sand Properties	
Sand Diameter (microns)	500
Tube Diameter (in)	4
Small Flow Rate (LPM)	36.09
Large Flow Rate (LPM)	426.8
Area (m²)	8.11E-03
Sphericity	0.8
Viscosity of Air (Pa s)	1.82E-05
Density of air (kg/m³)	1.184
Density of particle (kg/m³)	1.28E+03
Porosity	0.49
Gravity (m/s²)	9.8
Ergun Min Fluidization Velocity (m/s)	1.69E-01
Actual Min Fluidization Velocity (m/s)	2.19E-01

Tables 7.2: Pressure Drop, Bed Height Data Collected, and Velocity Calculated for 6in. Silica Sand

	Packing Height (cm)	P1	P2	ΔP (in. H ₂ O)	ΔP (Pa)	% of 426.8 LPM	Flow Rate LPM	% of 36.09 LPM	Flow Rate LPM	Flow Rate (m ³ /s)	Velocity (m/s)	Bed Height with gas (cm)	Packing Height (cm)
Increasing Pressure	15.24	3.7	2.7	1	248.84			20	7.218	0.00012	1.48E-02	4.334928	15.24
	15.24	4.3	2.2	2.1	522.564			40	14.436	0.000241	2.97E-02	4.334928	15.24
	15.24	4.9	1.6	3.3	821.172			60	21.654	0.000361	4.45E-02	4.334928	15.24
	15.24	5.45	1.1	4.35	1082.454			80	28.872	0.000481	5.94E-02	4.334928	15.24
	15.24	5.9	0.6	5.3	1318.852			100	36.09	0.000602	7.42E-02	4.334928	15.24
	15.24	5.5	0.6	4.9	1219.316	15	64.02			0.001067	1.32E-01	4.334928	16.8
	15.24	5.7	0.7	5	1244.2	20	85.36			0.001423	1.75E-01	4.334928	18.3
	15.24	5.6	0.8	4.8	1194.432	25	106.7			0.001778	2.19E-01	4.334928	23.3
	15.24	5.6	0.9	4.7	1169.548	30	128.04			0.002134	2.63E-01	4.334928	26.7
	15.24	5.6	0.9	4.7	1169.548	35	149.38			0.00249	3.07E-01	4.334928	29.6
	15.24	5.6	0.9	4.7	1169.548	40	170.72			0.002845	3.51E-01	4.334928	31.4
Decreasing Pressure	15.24	3.6	2.8	0.8	199.072			20	7.218	0.00012	1.48E-02	4.334928	15.9
	15.24	4	2.4	1.6	398.144			40	14.436	0.000241	2.97E-02	4.334928	15.9
	15.24	4.4	2	2.4	597.216			60	21.654	0.000361	4.45E-02	4.334928	15.9
	15.24	4.9	1.6	3.3	821.172			80	28.872	0.000481	5.94E-02	4.334928	16
	15.24	5.3	1.1	4.2	1045.128			100	36.09	0.000602	7.42E-02	4.334928	16.3
	15.24	5.5	1	4.5	1119.78	15	64.02			0.001067	1.32E-01	4.334928	16.6
	15.24	5.5	1	4.5	1119.78	20	85.36			0.001423	1.75E-01	4.334928	18.3
	15.24	5.6	0.9	4.7	1169.548	25	106.7			0.001778	2.19E-01	4.334928	20.2
	15.24	5.6	0.9	4.7	1169.548	30	128.04			0.002134	2.63E-01	4.334928	24.4
	15.24	5.6	0.9	4.7	1169.548	35	149.38			0.00249	3.07E-01	4.334928	28.2
	15.24	5.6	0.9	4.7	1169.548	40	170.72			0.002845	3.51E-01	4.334928	31.4

Table 7.3: Ergun Equation Data for Pressure Drop for 6in. Silica Sand

Predicted Pressure Drop (Pa)	A	B	C	D
8.6E+01	2.5E+02	2.210814	1.14E+00	4.334928
1.7E+02	5.1E+02	2.210814	4.56E+00	4.334928
2.6E+02	7.6E+02	2.210814	1.03E+01	4.334928
3.5E+02	1.0E+03	2.210814	1.82E+01	4.334928
4.5E+02	1.3E+03	2.210814	2.85E+01	4.334928
9.0E+02	2.2E+03	2.210814	8.97E+01	4.334928
1.3E+03	3.0E+03	2.210814	1.60E+02	4.334928
2.2E+03	3.7E+03	2.210814	2.49E+02	4.334928
3.1E+03	4.5E+03	2.210814	3.59E+02	4.334928
4.1E+03	5.2E+03	2.210814	4.88E+02	4.334928
5.0E+03	6.0E+03	2.210814	6.38E+02	4.334928
9.0E+01	2.5E+02	2.210814	1.14E+00	4.334928
1.8E+02	5.1E+02	2.210814	4.56E+00	4.334928
2.7E+02	7.6E+02	2.210814	1.03E+01	4.334928
3.7E+02	1.0E+03	2.210814	1.82E+01	4.334928
4.8E+02	1.3E+03	2.210814	2.85E+01	4.334928
8.9E+02	2.2E+03	2.210814	8.97E+01	4.334928
1.3E+03	3.0E+03	2.210814	1.60E+02	4.334928
1.9E+03	3.7E+03	2.210814	2.49E+02	4.334928
2.8E+03	4.5E+03	2.210814	3.59E+02	4.334928
3.9E+03	5.2E+03	2.210814	4.88E+02	4.334928
5.0E+03	6.0E+03	2.210814	6.38E+02	4.334928

Tables 7.4: Pressure Drop, Bed Height Data Collected, and Velocity Calculated for 10in. Silica Sand

	Packing Height (cm)	P1	P2	ΔP (in. H ₂ O)	ΔP (Pa)	% of 426.8 LPM	Flow Rate LPM	% of 36.09 LPM	Flow Rate LPM	Flow Rate (m ³ /s)	Velocity (m/s)	Bed Height with gas (cm)
Increasing Pressure	25.4	3.8	2.5	1.3	323.492			20	7.218	0.00012	1.48E-02	25
	25.4	4.6	1.7	2.9	721.636			40	14.436	0.000241	2.97E-02	25
	25.4	5.4	1	4.4	1094.896			60	21.654	0.000361	4.45E-02	25
	25.4	6.2	0.1	6.1	1517.924			80	28.872	0.000481	5.94E-02	25
	25.4	7	-0.6	7.6	1891.184			100	36.09	0.000602	7.42E-02	25
	25.4	7.1	-0.8	7.9	1965.836	15	64.02			0.001067	1.32E-01	25
	25.4	7.5	-1.1	8.6	2140.024	20	85.36			0.001423	1.75E-01	26.3
	25.4	7.4	-0.9	8.3	2065.372	25	106.7			0.001778	2.19E-01	30.1
	25.4	7.4	-0.9	8.3	2065.372	30	128.04			0.002134	2.63E-01	32.1
	25.4	7.4	-0.9	8.3	2065.372	35	149.38			0.00249	3.07E-01	35.2
Decreasing Pressure	25.4	3.8	2.7	1.1	273.724			20	7.218	0.00012	1.48E-02	24.6
	25.4	4.4	2	2.4	597.216			40	14.436	0.000241	2.97E-02	24.6
	25.4	5	1.4	3.6	895.824			60	21.654	0.000361	4.45E-02	24.6
	25.4	5.7	0.7	5	1244.2			80	28.872	0.000481	5.94E-02	24.6
	25.4	6.5	-0.1	6.6	1642.344			100	36.09	0.000602	7.42E-02	24.7
	25.4	6.7	-0.4	7.1	1766.764	15	64.02			0.001067	1.32E-01	25.2
	25.4	7.1	-0.7	7.8	1940.952	20	85.36			0.001423	1.75E-01	27.3
	25.4	7.3	-0.8	8.1	2015.604	25	106.7			0.001778	2.19E-01	29.6
	25.4	7.3	-0.8	8.1	2015.604	30	128.04			0.002134	2.63E-01	31.9
	25.4	7.4	-0.9	8.3	2065.372	35	149.38			0.00249	3.07E-01	35.2
	25.4	7.4	-0.9	8.3	2065.372	40	170.72			0.002845	3.51E-01	39.6

Table 7.5: Ergun Equation Data for Pressure Drop for 10in. Silica Sand

Predicted Pressure Drop (Pa)	A	B	C	D
1.4E+02	2.5E+02	2.210814	1.14E+00	4.334928
2.9E+02	5.1E+02	2.210814	4.56E+00	4.334928
4.3E+02	7.6E+02	2.210814	1.03E+01	4.334928
5.8E+02	1.0E+03	2.210814	1.82E+01	4.334928
7.3E+02	1.3E+03	2.210814	2.85E+01	4.334928
1.3E+03	2.2E+03	2.210814	8.97E+01	4.334928
1.9E+03	3.0E+03	2.210814	1.60E+02	4.334928
2.8E+03	3.7E+03	2.210814	2.49E+02	4.334928
3.7E+03	4.5E+03	2.210814	3.59E+02	4.334928
4.8E+03	5.2E+03	2.210814	4.88E+02	4.334928
6.3E+03	6.0E+03	2.210814	6.38E+02	4.334928
1.4E+02	2.5E+02	2.210814	1.14E+00	4.334928
2.8E+02	5.1E+02	2.210814	4.56E+00	4.334928
4.2E+02	7.6E+02	2.210814	1.03E+01	4.334928
5.7E+02	1.0E+03	2.210814	1.82E+01	4.334928
7.2E+02	1.3E+03	2.210814	2.85E+01	4.334928
1.4E+03	2.2E+03	2.210814	8.97E+01	4.334928
2.0E+03	3.0E+03	2.210814	1.60E+02	4.334928
2.8E+03	3.7E+03	2.210814	2.49E+02	4.334928
3.7E+03	4.5E+03	2.210814	3.59E+02	4.334928
4.8E+03	5.2E+03	2.210814	4.88E+02	4.334928
6.3E+03	6.0E+03	2.210814	6.38E+02	4.334928

Table 7.6: Ion Exchange Resin Properties

Resin Properties	
Resin Diameter (microns)	600
Tube Diameter (in)	4
Small Flow Rate (LPM)	36.09
Large Flow Rate (LPM)	426.8
Area (m ²)	8.11E-03
Sphericity	1
Viscosity of Air (Pa s)	1.82E-05
Density of air (kg/m ³)	1.184
Density of particle (kg/m ³)	7.00E+02
Porosity	0.523
Gravity (m/s ²)	9.8
Ergun Min Fluidization Velocity (m/s)	2.71E-01
Actual Min Fluidization Velocity (m/s)	2.19E-01

Tables 7.7: Pressure Drop, Bed Height Data Collected, and Velocity Calculated for 6in. Ion Exchange Resin

	Packing Height (cm)	P1	P2	ΔP (in. H ₂ O)	ΔP (Pa)	% of 426.8 LPM	Flow Rate LPM	% of 36.09 LPM	Flow Rate LPM	Flow Rate (m ³ /s)	Velocity (m/s)	Bed Height with gas (cm)
Increasing pressure	15.24	3.4	3	0.4	99.536			20	7.218	0.00012	1.48E-02	15.2
	15.24	3.6	2.8	0.8	199.072			40	14.436	0.000241	2.97E-02	15.2
	15.24	3.9	2.6	1.3	323.492			60	21.654	0.000361	4.45E-02	15.2
	15.24	4.1	2.4	1.7	423.028			80	28.872	0.000481	5.93E-02	15.2
	15.24	4.4	2.1	2.3	572.332			100	36.09	0.000602	7.42E-02	15.2
	15.24	4.5	1.9	2.6	646.984	15	64.02			0.001067	1.32E-01	15.2
	15.24	4.8	1.7	3.1	771.404	20	85.36			0.001423	1.75E-01	15.6
	15.24	4.8	1.6	3.2	796.288	25	106.7			0.001778	2.19E-01	17
	15.24	4.8	1.6	3.2	796.288	30	128.04			0.002134	2.63E-01	18.3
	15.24	4.8	1.6	3.2	796.288	35	149.38			0.00249	3.07E-01	20.1
	15.24	4.8	1.6	3.2	796.288	40	170.72			0.002845	3.51E-01	21.6
Decreasing Pressure	15.24	3.4	3.1	0.3	74.652			20	7.218	0.00012	1.48E-02	14.9
	15.24	3.6	2.9	0.7	174.188			40	14.436	0.000241	2.97E-02	14.9
	15.24	3.8	2.7	1.1	273.724			60	21.654	0.000361	4.45E-02	14.9
	15.24	3.9	2.5	1.4	348.376			80	28.872	0.000481	5.93E-02	14.9
	15.24	4.2	2.2	2	497.68			100	36.09	0.000602	7.42E-02	14.9
	15.24	4.5	1.9	2.6	646.984	15	64.02			0.001067	1.32E-01	14.9
	15.24	4.8	1.6	3.2	796.288	20	85.36			0.001423	1.75E-01	15.4
	15.24	4.8	1.6	3.2	796.288	25	106.7			0.001778	2.19E-01	16.3
	15.24	4.8	1.6	3.2	796.288	30	128.04			0.002134	2.63E-01	18.2
	15.24	4.8	1.6	3.2	796.288	35	149.38			0.00249	3.07E-01	19.8
	15.24	4.8	1.6	3.2	796.288	40	170.72			0.002845	3.51E-01	21.6

Table 7.8: Ergun Equation Data for Pressure Drop for 6in. Ion Exchange Resin

Predicted Pressure Drop (Pa)	A	B	C	D
2.8E+01	1.1E+02	1.590493	7.60E-01	3.334366
5.6E+01	2.2E+02	1.590493	3.04E+00	3.334366
8.5E+01	3.4E+02	1.590493	6.84E+00	3.334366
1.1E+02	4.5E+02	1.590493	1.22E+01	3.334366
1.5E+02	5.6E+02	1.590493	1.90E+01	3.334366
2.7E+02	1.0E+03	1.590493	5.98E+01	3.334366
3.9E+02	1.3E+03	1.590493	1.06E+02	3.334366
5.4E+02	1.7E+03	1.590493	1.66E+02	3.334366
7.3E+02	2.0E+03	1.590493	2.39E+02	3.334366
9.6E+02	2.3E+03	1.590493	3.25E+02	3.334366
1.2E+03	2.7E+03	1.590493	4.25E+02	3.334366
2.7E+01	1.1E+02	1.590493	7.60E-01	3.334366
5.5E+01	2.2E+02	1.590493	3.04E+00	3.334366
8.3E+01	3.4E+02	1.590493	6.84E+00	3.334366
1.1E+02	4.5E+02	1.590493	1.22E+01	3.334366
1.4E+02	5.6E+02	1.590493	1.90E+01	3.334366
2.7E+02	1.0E+03	1.590493	5.98E+01	3.334366
3.8E+02	1.3E+03	1.590493	1.06E+02	3.334366
5.2E+02	1.7E+03	1.590493	1.66E+02	3.334366
7.2E+02	2.0E+03	1.590493	2.39E+02	3.334366
9.5E+02	2.3E+03	1.590493	3.25E+02	3.334366
1.2E+03	2.7E+03	1.590493	4.25E+02	3.334366

Tables 7.9: Pressure Drop, Bed Height Data Collected, and Velocity Calculated for 10in. Ion Exchange Resin

	Packing Height (cm)	P1	P2	ΔP (in. H ₂ O)	ΔP (Pa)	% of 426.8 LPM	Flow Rate LPM	% of 36.09 LPM	Flow Rate LPM	Flow Rate (m ³ /s)	Velocity (m/s)	Bed Height with gas (cm)
Increasing Pressure	25.4	3.5	2.9	0.6	149.304			20	7.218	0.0001203	1.48E-02	25.3
	25.4	3.8	2.6	1.2	298.608			40	14.436	0.0002406	2.97E-02	25.3
	25.4	4.2	2.3	1.9	472.796			60	21.654	0.0003609	4.45E-02	25.3
	25.4	4.6	1.8	2.8	696.752			80	28.872	0.0004812	5.94E-02	25.3
	25.4	4.9	1.5	3.4	846.056			100	36.09	0.0006015	7.42E-02	25.3
	25.4	5.3	1.2	4.1	1020.244	15	64.02			0.001067	1.32E-01	25.2
	25.4	6.1	0.3	5.8	1443.272	20	85.36			0.001422667	1.75E-01	25.2
	25.4	6.3	0.1	6.2	1542.808	25	106.7			0.001778333	2.19E-01	26.6
	25.4	6.3	0.1	6.2	1542.808	30	128.04			0.002134	2.63E-01	28.1
	25.4	6.3	0.1	6.2	1542.808	35	149.38			0.002489667	3.07E-01	29.6
	25.4	6.3	0.1	6.2	1542.808	40	170.72			0.002845333	3.51E-01	31.4
Decreasing Pressure	25.4	3.5	3	0.5	124.42			20	7.218	0.0001203	1.48E-02	25
	25.4	3.8	2.6	1.2	298.608			40	14.436	0.0002406	2.97E-02	25
	25.4	4.2	2.3	1.9	472.796			60	21.654	0.0003609	4.45E-02	25
	25.4	4.5	1.9	2.6	646.984			80	28.872	0.0004812	5.94E-02	25
	25.4	4.9	1.5	3.4	846.056			100	36.09	0.0006015	7.42E-02	25
	25.4	5.3	1.1	4.2	1045.128	15	64.02			0.001067	1.32E-01	25
	25.4	6.1	0.4	5.7	1418.388	20	85.36			0.001422667	1.75E-01	25
	25.4	6.3	0.1	6.2	1542.808	25	106.7			0.001778333	2.19E-01	26.2
	25.4	6.3	0.1	6.2	1542.808	30	128.04			0.002134	2.63E-01	27.8
	25.4	6.3	0.1	6.2	1542.808	35	149.38			0.002489667	3.07E-01	29.7
	25.4	6.3	0.1	6.2	1542.808	40	170.72			0.002845333	3.51E-01	31.4

Table 7.10: Ergun Equation Data for Pressure Drop for 10in. Ion Exchange Resin

Predicted Pressure Drop (Pa)	A	B	C	D
4.6E+01	1.1E+02	1.590493	7.60E-01	3.334366
9.3E+01	2.3E+02	1.590493	3.04E+00	3.334366
1.4E+02	3.4E+02	1.590493	6.84E+00	3.334366
1.9E+02	4.5E+02	1.590493	1.22E+01	3.334366
2.4E+02	5.6E+02	1.590493	1.90E+01	3.334366
4.5E+02	1.0E+03	1.590493	5.98E+01	3.334366
6.2E+02	1.3E+03	1.590493	1.06E+02	3.334366
8.5E+02	1.7E+03	1.590493	1.66E+02	3.334366
1.1E+03	2.0E+03	1.590493	2.39E+02	3.334366
1.4E+03	2.3E+03	1.590493	3.26E+02	3.334366
1.8E+03	2.7E+03	1.590493	4.25E+02	3.334366
4.5E+01	1.1E+02	1.590493	7.60E-01	3.334366
9.2E+01	2.3E+02	1.590493	3.04E+00	3.334366
1.4E+02	3.4E+02	1.590493	6.84E+00	3.334366
1.9E+02	4.5E+02	1.590493	1.22E+01	3.334366
2.4E+02	5.6E+02	1.590493	1.90E+01	3.334366
4.5E+02	1.0E+03	1.590493	5.98E+01	3.334366
6.2E+02	1.3E+03	1.590493	1.06E+02	3.334366
8.4E+02	1.7E+03	1.590493	1.66E+02	3.334366
1.1E+03	2.0E+03	1.590493	2.39E+02	3.334366
1.4E+03	2.3E+03	1.590493	3.26E+02	3.334366
1.8E+03	2.7E+03	1.590493	4.25E+02	3.334366

Table 7.11: Ion Exchange Resin Porosity

	Unit	
Ion Exchange Resin	mL	10
Water-Initial	mL	50
Water-Final	mL	39
Porosity Calculated by Saturation		0.523

Table 7.12: Silica Sand Porosity

	Unit	
Sand	mL	9.5
Water-Initial	mL	49
Water-Final	mL	40
Porosity Calculated by Saturation		0.49

7.13: Particle Size Data and Average

Particle Size (μm)	Mass of particle size with Beaker (g)	Mass of Particle Size without Beaker (g)	Mass Fraction of particle size.	$\frac{x_i}{P_{di}}$
595	52.5	20.7	0.1401	2.36E-04
420	88.8	57.0	0.3859	9.19E-04
210	87.3	55.5	0.3758	1.79E-03
105	46.3	14.5	0.0982	9.35E-04
Avg Particle Size (μm)	257.8178173	$= \frac{1}{\sum \frac{x_i}{P_{di}}}$		

8. Appendix II: Error Analysis

There were numerous sources of error for this experiment. The environment wasn't controlled, starting with varying the flow rate from two different valves caused discrepancies. The bed itself had cracks at the inlet, this resulted in pressure exiting the system. The equipment added an error factor to the results because of the uncertainty from reading of pressure drop. There was also error in determining the height of the bed, which made it difficult to obtain a single reading since the height kept changing. Another error could have been from measuring the porosities of the Silica Sand and Ion Exchange Resin, the volumes of the water and solid could only be estimated to the tenth's place. The two types of particles were vacuumed with the same vacuum, so the particles could've been contaminated. The experimental error for the Silica Sand was 29.57% and for the Ion Exchange Resin was 19.04%. Such errors could be from the apparatus being older and the equipment failing to provide accurate readings.

Table 8.1: Error Analysis

Minimum Fluidization Velocities m/s		
Data	Silica Sand	Resin Beads
Experimental	0.219349108	0.219276613
Ergun Equation	0.169290945	0.270841654
Percent Error	29.56930875	19.03881468

9. Appendix III: Sample Calculations

Percent Error

$$\frac{(\text{Actual} - \text{Theoretical})}{\text{Theoretical}} * 100 = \text{percent error}$$
$$\frac{0.219349108 - 0.169290945}{0.169290945} * 100 = 29.56\%$$

Porosity

$$\text{Bed Voidage} = \frac{\text{Volume of water added}}{\text{Volume of water added} + \text{Volume of Resin}}$$

$$\text{Porosity of Resin} = \frac{11\text{ml}}{11\text{ml} + 10\text{ml}} = 0.523$$

Cross Sectional Area

$$\pi \left(\frac{D}{2} \right)^2$$
$$= \pi \left(\frac{4\text{in} \cdot 0.0254 \frac{\text{m}}{\text{in}}}{2} \right)^2 = 0.00811 \text{ m}^2$$

Minimum Fluidization (Sand)

$$V_{om} = \frac{g(\rho_P - \rho)}{150\mu} \frac{\varepsilon_M^3}{1 - \varepsilon_M} \phi_s^2 D_p^2$$
$$V_{om} = \frac{9.8 (.000128 - 1.184)}{150 * .0000182} \frac{0.49^3}{1 - 0.49} * 0.8^2 * \left(\frac{500}{1000000} \right)^2 = 0.169 \text{ m/s}$$

Ergun Equation-Predicted Pressure (Bed Height: 6in Silica Sand, 20 % flow @ 36.09LPM)

$$\frac{\Delta P}{L} = \frac{150 \bar{V}_0 \mu}{\Phi_s D_p} * \frac{1 - \varepsilon^2}{\varepsilon^3} + \frac{1.75 \rho V_0^2}{\phi_s D_p} \frac{1 - \varepsilon}{\varepsilon^3}$$

$$\Delta P = \left(\frac{150 \bar{V}_0 \mu}{\Phi_s D_p} * \frac{1 - \varepsilon^2}{\varepsilon^3} + \frac{1.75 \rho \bar{V}_0^2}{\Phi_s D_p} \frac{1 - \varepsilon}{\varepsilon^3} \right) * L$$

$$\left(\frac{150 * (0.0148) * (0.0000182) * (1 - 0.49)^2}{0.8 * \left(\frac{500}{1000000} \right) (0.49)^3} + \frac{1.75 * (1.184) * (0.0148)^2 * 1 - 0.49}{0.8 * \left(\frac{500}{1000000} \right) 0.49^3} \right) * \left(\frac{15.24}{100} \right)$$

$$= 0.086 \text{ Pa}$$

10. Appendix IV: Individual Team Contributions

NAME: Anne, Evelyn, Maurice, Sandra and Tianna

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	All
Pre-Lab Editing	2.5	All
Experimental Objective (Prelab)	0.5	Anne
Apparatus (prelab)	1	Sandra & Maurice
Materials & Supplies (prelab)	0.5	Maurice
Procedure (prelab)	1	Tianna
Project Safety Evaluation (prelab)	1	Evelyn
Final Lab Editing	0.5	All
Executive Summary (final lab)	2	Sandra
Introduction (Final lab)	2	Tianna
Theory (Final Lab)	2	Evelyn & Tianna
Results (final lab)	5.5	Sandra & Evelyn
Discussion (final lab)	3	Maurice
References (final and/or prelab)	0.5	Tianna
Data Tabulation/Graphs (final)	4	Sandra & Evelyn
Error Analysis (final lab)	3.5	Maurice, Anne & Evelyn
Sample Calculations (final Lab)	2	Evelyn
Power Point Presentation	3	Maurice & Anne
Total	38.5	